

6 Manganese

Manganese supports photosynthesis and enzyme activity. Its availability can change sharply with pH, drainage, and oxidation conditions in the root zone.

Why it matters

- **Photosynthesis and enzymes.** Manganese participates in water splitting during photosynthesis and activates many enzymes.
- **Strong environmental response.** The same soil may shift between low and excessive soluble Mn as pH and wetness change.

Tentative leaf guide

Measure	Interpretation note
Tentative avocado range	UC guidance lists about 10 to 15 parts per million as deficient, 30 to 500 ppm as adequate, and more than 1000 ppm as excess.
Sampling caution	Dust, sprays, leaf age, and inconsistent sampling can distort trace-element results.

Main soil and environmental controls

Factor	What it can mean in the root zone
pH	High pH usually lowers Mn availability. Strong acidity increases solubility and can raise toxicity risk.
Waterlogging and redox	Low-oxygen reducing conditions can convert Mn to more soluble forms. A high tissue value may therefore signal wet or poorly aerated soil.
Organic matter and microbes	Decomposition changes oxygen and redox conditions and produces compounds that can bind or mobilize Mn.
Texture and salinity	Fine or compacted soil may remain wet longer. Salinity and sodium can reduce root activity and damage structure.
Nutrient context	Mn often changes with Fe, Zn, Mg, Ca, and root condition; isolated interpretation is weak.

Field clues and checks

- **Read the leaf pattern carefully.** Deficiency often appears as interveinal chlorosis, usually with wider green bands along veins than classic iron chlorosis.
- **Inspect wet areas.** Compare Mn, root color, smell, soil moisture, pH, EC, and drainage between affected and healthy zones.
- **Avoid reflex spraying.** A leaf result may be reporting a pH or oxygen problem rather than a shortage that added Mn will correct.

DST interpretation Phase 1 found that Mn moved with Mg, Zn, Fe, and Ca and did not show a stable adjusted yield effect. Treat Mn as a possible root-zone or nutrient-state marker, then inspect pH, redox, irrigation, salinity, organic matter, and roots.